PRODUCT REQUIREMENTS DOCUMENT (PRD)

Multi-Agent Cyber Defense System Using Agentic AI

Version: 1.0

Prepared By: <*Your Name*>

Department: Computer Science and Engineering

Institution: <*College Name*>

Academic Year: 2026–2027

Project Type: Artificial Intelligence Final Year Project

Table of Contents

1. Executive Summary
2. Background
3. Problem Statement
4. Project Vision
5. Business Objectives
6. Product Scope
7. Stakeholders
8. System Overview
9. Multi-Agent Architecture
10. Functional Requirements
11. Non-Functional Requirements
12. AI Algorithms
13. Cybersecurity Framework
14. System Workflow
15. Database Design
16. API Design
17. Dashboard Requirements
18. Testing Strategy
19. Deployment
20. Risk Assessment
21. Future Enhancements
22. Appendix

---

1 Executive Summary

1.1 Introduction

Cybersecurity has become one of the most critical challenges in the digital era. Organizations continuously face sophisticated cyberattacks including ransomware, phishing, malware, denial-of-service attacks, insider threats, credential theft, and advanced persistent threats. Traditional cybersecurity solutions primarily depend on signature-based detection systems, manually configured rules, or isolated machine learning models. While these approaches remain valuable, they often struggle to adapt to rapidly evolving attack techniques and frequently require significant human intervention.

Recent advances in Artificial Intelligence have introduced a new paradigm known as **Agentic AI**, where autonomous intelligent agents collaborate to achieve complex objectives. Unlike conventional AI systems that perform isolated prediction tasks, Agentic AI systems perceive their environment, reason about observations, communicate with other agents, plan actions, execute decisions, and continuously improve through feedback.

This project proposes a **Multi-Agent Cyber Defense System (MACDS)** that applies Agentic AI principles to cybersecurity. The system consists of multiple specialized AI agents working collaboratively to monitor network traffic, detect threats, analyze attack behavior, reason about security events, plan appropriate responses, and automate defensive actions. Each agent has clearly defined responsibilities while sharing information through a secure communication framework.

The proposed architecture integrates classical Artificial Intelligence concepts—including intelligent agents, search algorithms, game theory, constraint satisfaction problems, knowledge representation, propositional logic, and first-order logic—with modern cybersecurity practices. By combining these techniques, the system aims to improve threat detection accuracy, reduce response times, and enhance explainability in automated cybersecurity operations.

1.2 Purpose of the Product

The primary purpose of the Multi-Agent Cyber Defense System is to create an intelligent cybersecurity platform capable of autonomously detecting, investigating, and mitigating cyber threats with minimal human intervention. The product is designed as both an educational demonstration of Artificial Intelligence concepts and a practical prototype for modern cyber defense.

The system seeks to:

- Continuously monitor network activity.
- Identify malicious behavior using intelligent search and reasoning algorithms.
- Coordinate multiple AI agents for collaborative decision-making.
- Recommend or execute automated defensive actions.

- Provide transparent explanations for every security decision.
- Demonstrate the practical application of Artificial Intelligence concepts covered in the academic curriculum.

1.3 Intended Audience

This Product Requirements Document is intended for:

- Project supervisors
- Internal project developers
- Artificial Intelligence researchers
- Cybersecurity researchers
- Software architects
- Quality assurance engineers
- Final-year project evaluators
- Academic reviewers

---

2 Background

2.1 Current Cybersecurity Landscape

The rapid adoption of cloud computing, Internet of Things (IoT), mobile technologies, and distributed systems has significantly expanded the attack surface of modern organizations. Cybercriminals now employ automated attack tools capable of exploiting vulnerabilities at unprecedented speed.

Security Operations Centers (SOCs) receive thousands of alerts every day. Many of these alerts are false positives, forcing analysts to spend valuable time investigating benign events. Manual analysis cannot keep pace with modern attack volumes.

Furthermore, sophisticated attacks often involve multiple stages:

- Initial reconnaissance
- Vulnerability scanning
- Credential compromise
- Lateral movement
- Privilege escalation
- Data exfiltration
- Persistence establishment

Traditional security tools usually detect individual events rather than understanding the entire attack chain.

2.2 Limitations of Existing Solutions

Although current cybersecurity technologies provide significant protection, several limitations remain.

Rule-Based Systems

Rule-based Intrusion Detection Systems depend heavily on manually written detection signatures. Unknown attacks and zero-day exploits frequently bypass these systems.

Machine Learning Models

Machine learning models often function as "black boxes," making it difficult for analysts to understand why an alert was generated. They also require large, high-quality datasets and periodic retraining.

Human-Centric Investigation

Many organizations rely on human analysts to investigate alerts. This approach becomes impractical as event volumes continue to grow.

Isolated Security Components

Firewalls, antivirus software, intrusion detection systems, and endpoint protection platforms often operate independently with limited coordination.

Lack of Adaptive Decision Making

Most security products detect attacks but cannot independently determine the best response strategy.

2.3 Why Agentic AI?

Agentic AI represents the next evolution of intelligent software systems.

Unlike traditional AI models that focus on prediction, Agentic AI systems:

- Observe their environment.
- Form goals.
- Plan actions.
- Execute decisions.
- Learn from outcomes.
- Collaborate with other agents.

This makes Agentic AI particularly suitable for cybersecurity, where rapid decision-making and collaboration are essential.

3 Problem Statement

Modern organizations face increasingly sophisticated cyber threats while relying on fragmented security tools that require significant manual intervention. Existing security solutions often produce excessive false positives, lack contextual understanding, and struggle to coordinate responses across multiple systems.

There is a growing need for an autonomous cybersecurity platform capable of continuously monitoring network activity, intelligently analyzing threats, collaborating across specialized agents, reasoning about attack scenarios, and automatically responding to incidents while providing transparent explanations for its decisions.

The proposed Multi-Agent Cyber Defense System addresses these challenges by leveraging Agentic AI to build an intelligent, explainable, and collaborative cybersecurity framework.

---

4 Project Vision

To develop an autonomous, explainable, and scalable cybersecurity platform that combines classical Artificial Intelligence algorithms with modern Agentic AI principles to proactively defend enterprise networks against evolving cyber threats.

The long-term vision is to create a modular architecture in which specialized intelligent agents collaborate seamlessly, enabling organizations to reduce incident response times, improve threat detection accuracy, and enhance operational efficiency.

---

5 Business Objectives

The project aims to achieve the following business objectives:

1. Reduce the average time required to detect cyber threats.
2. Minimize false-positive security alerts.
3. Improve collaboration between autonomous security components.
4. Enhance transparency through explainable AI decisions.
5. Demonstrate practical applications of Artificial Intelligence concepts.
6. Provide a scalable architecture that supports future expansion.
7. Enable automated incident response for common attack scenarios.
8. Establish a knowledge base that continuously improves security intelligence.

---

6 Product Scope

6.1 In Scope

The following capabilities are included in the project:

- Autonomous network monitoring
- Multi-agent communication
- Threat detection
- Logical reasoning
- Knowledge-based inference
- Search algorithm implementation
- Game-theoretic defense planning
- Constraint satisfaction for policy validation
- Dashboard visualization
- Alert management
- Incident reporting
- Rule-based reasoning
- Attack classification
- Automated response recommendations
- Security event logging
- Explainable AI outputs

6.2 Out of Scope

The following features are intentionally excluded from the initial release:

- Commercial antivirus replacement
- Real-time protection for enterprise-scale networks
- Full SIEM implementation
- Cloud-native distributed deployment
- Deep packet inspection at carrier scale
- Large language model training
- Hardware firewall integration
- Commercial SOC replacement

7. Stakeholder Analysis

7.1 Overview

The success of the Multi-Agent Cyber Defense System depends on collaboration among several stakeholders. Each stakeholder interacts with the system differently and has unique functional requirements.

Stakeholder	Role	Responsibilities
System Administrator	Primary User	Configure agents, monitor threats, manage users
Security Analyst	Investigator	Analyze alerts, validate incidents, review reports
Network Administrator	Infrastructure Manager	Maintain network topology and firewall policies
AI Agents	Autonomous Actors	Detect, reason, collaborate, and respond to threats
Database Administrator	Data Manager	Manage security logs, rules, and incident database
Project Administrator	System Owner	Configure policies and overall system settings

---

7.2 Stakeholder Responsibilities

System Administrator

Responsibilities include:

- User management
- Agent configuration
- Dashboard monitoring
- Security policy updates
- Reviewing system health
- Responding to escalated incidents

Security Analyst

Responsibilities include:

- Reviewing detected threats
- Validating alerts
- Approving automated responses (when required)
- Updating detection rules
- Investigating attack chains

Network Administrator

Responsibilities include:

- Firewall management
- VLAN configuration

- Network segmentation
- Device onboarding
- Traffic monitoring

AI Agents

Unlike human stakeholders, AI agents act autonomously.

Responsibilities include:

- Perceiving network events
- Communicating findings
- Sharing knowledge
- Planning actions
- Executing responses
- Learning from outcomes

---

8. Product Overview

8.1 Product Description

The Multi-Agent Cyber Defense System is an autonomous cybersecurity platform composed of specialized software agents that cooperate to secure computer networks.

Instead of relying on a single intelligent module, responsibilities are divided among independent AI agents. Each agent possesses:

- Defined goals
- Environmental awareness
- Decision-making capability
- Communication interfaces
- Knowledge access
- Autonomous execution capability

Together these agents continuously monitor network activity and collaborate to identify, analyze, and mitigate cyber threats.

8.2 High-Level System Architecture

---

9. Multi-Agent Architecture

9.1 Why Multi-Agent Architecture?

Cybersecurity problems are naturally distributed.

Different security tasks require different expertise:

- Monitoring
- Detection
- Investigation
- Decision making
- Response
- Reporting

Instead of building one enormous AI model, this project divides responsibilities among intelligent autonomous agents.

Benefits include:

- Higher scalability
- Parallel execution
- Easier maintenance
- Better explainability
- Independent upgrades
- Fault isolation

9.2 Agent Hierarchy

The Coordinator Agent manages workflow and ensures the correct sequence of operations.

---

10. Detailed Agent Design

10.1 Coordinator Agent

Purpose

Acts as the orchestrator of the entire system.

It assigns tasks to other agents and manages communication.

Responsibilities

- Register agents
- Schedule tasks
- Monitor agent health
- Handle failures
- Coordinate investigations
- Resolve task conflicts

Inputs

- Agent messages
- System events
- Incident reports

Outputs

- Task assignments
- Coordination messages
- Incident workflow

AI Concepts

- Intelligent Agent
- Rational Agent
- Utility-Based Decision Making

10.2 Network Monitoring Agent

Purpose

Continuously monitors network activity.

This agent represents the system's perception layer.

Responsibilities

- Capture packets
- Monitor traffic
- Monitor bandwidth
- Detect abnormal traffic volume
- Observe active devices

Inputs

- Network packets
- Switch logs
- Router logs
- NetFlow records

Outputs

- Suspicious events
- Traffic statistics
- Device activity reports

AI Mapping

Unit I

- Intelligent Agent
- Rationality
- Environment Perception

Search Algorithms

- Breadth First Search
- Depth First Search
- Online Search

10.3 Intrusion Detection Agent

Purpose

Determines whether observed activity represents malicious behavior.

Detects

- Port scanning
- SQL Injection
- Brute force attacks
- Malware communication
- Command and Control traffic
- DDoS attacks
- DNS tunneling

Inputs

- Traffic metadata
- Packet analysis
- Monitoring alerts

Outputs

- Threat score
- Attack category
- Confidence value

Algorithms

- A*
- Greedy Best First Search
- Heuristic Search

AI Mapping

Unit II

Informed Search

Heuristic Search

10.4 Threat Correlation Agent

Purpose

Individual alerts often appear harmless.

The Threat Correlation Agent combines multiple events into a single attack narrative.

Example

Responsibilities

- Event correlation
- Timeline creation
- Attack reconstruction
- Campaign identification

10.5 Knowledge Base Agent

Purpose

Stores cybersecurity knowledge.

Knowledge Includes

- Attack signatures
- Detection rules
- Firewall rules
- MITRE ATT&CK mappings (optional enhancement)
- Incident history
- Security policies

Functions

Retrieve

Store

Update

Delete

Version Rules

Example Rule

10.6 Reasoning Agent

Purpose

Acts as the logical reasoning engine.

This agent transforms evidence into conclusions.

Responsibilities

- Apply logical inference
- Validate hypotheses
- Eliminate false positives
- Recommend conclusions

Artificial Intelligence Concepts

Unit IV

Knowledge Based Agent

Propositional Logic

First Order Logic

Forward Chaining

Backward Chaining

Resolution

Example Forward Chaining

Rule 1

Rule 2

Rule 3

Example Backward Chaining

Goal

Required Evidence

10.7 Planning Agent

Purpose

Determines the optimal response strategy.

Rather than immediately blocking every threat, the Planning Agent evaluates multiple defensive actions and selects the most appropriate response based on risk, confidence, and operational impact.

Possible Actions

- Block IP address
- Disable user account
- Isolate endpoint
- Increase monitoring
- Notify administrator
- Ignore false positive

Algorithms

A*

Hill Climbing

Local Search

Game Theory

Alpha-Beta Search

Monte Carlo Tree Search

Example Planning Tree

10.8 Response Agent

Purpose

Executes defensive actions.

Responsibilities

- Block malicious IP
- Disable compromised account
- Kill malicious process
- Quarantine device
- Update firewall rules
- Generate alerts

Outputs

- Incident response report
- Security notifications
- Audit logs

10.9 Learning Agent

Purpose

Improves the system over time.

The Learning Agent analyzes previous incidents to optimize detection accuracy and reduce false positives.

Stores

- Previous attacks
- Response effectiveness
- False positives
- False negatives
- Analyst feedback

Future Enhancements

- Reinforcement Learning
- Case-Based Reasoning
- Adaptive rule generation

---

11. Agent Communication Framework

Communication Principles

Each agent communicates asynchronously using structured JSON messages over REST APIs or a message broker (e.g., Redis Pub/Sub, RabbitMQ).

Sample Message

---

12. Agent Collaboration Workflow

Mapping to AI Syllabus

Unit	Topics Used in Project
Unit I	Intelligent Agents, Rational Agents, Environment Modeling, Problem-Solving Agents, BFS, DFS, Online Search
Unit II	A*, Greedy Best-First Search, Hill Climbing, Local Search, Heuristic Search
Unit III	Game Theory, Alpha–Beta Pruning, Monte Carlo Tree Search, Constraint Satisfaction Problems
Unit IV	Knowledge-Based Agents, Propositional Logic, First-Order Logic, Forward Chaining, Backward Chaining, Resolution

13. Functional Requirements

13.1 Introduction

Functional requirements define **what the system must do**. These requirements describe the behavior expected from the Multi-Agent Cyber Defense System under normal operating conditions.

Each requirement is uniquely identified for traceability throughout the software development lifecycle.

Module 1 – Authentication & User Management

FR-001 User Authentication

Description

The system shall authenticate users before granting access.

Priority

High

Actor

System Administrator

Inputs

- Username
- Password

Outputs

- Authentication Token
- Login Status

Acceptance Criteria

- Valid credentials allow login.
- Invalid credentials display an error.
- Passwords shall never be stored in plaintext.

FR-002 Role-Based Access Control (RBAC)

The system shall provide role-based permissions.

Supported roles include:

- Administrator
- Security Analyst
- Network Administrator
- Read-Only Auditor

Only administrators can modify AI agents.

FR-003 Session Management

The system shall:

- Create secure sessions
- Automatically expire inactive sessions
- Prevent simultaneous duplicate logins (optional)

FR-004 Password Management

The system shall:

- Allow password reset
- Enforce password complexity
- Hash passwords
- Prevent password reuse

Module 2 – Dashboard

FR-005 Security Dashboard

The system shall display:

- Active threats
- Threat severity
- Agent status

- Network activity
- Incident timeline

Dashboard refresh interval:

≤ 2 seconds.

FR-006 Agent Health Monitoring

The dashboard shall display:

- Agent status
- Last heartbeat
- CPU usage
- Memory usage
- Response latency

FR-007 Live Threat Feed

The dashboard shall update detected attacks in real time.

Each threat shall display:

- Threat ID
- Timestamp
- Severity
- Attack Type
- Confidence Score

Module 3 – Network Monitoring Agent

FR-008 Packet Monitoring

The Monitoring Agent shall continuously inspect incoming network traffic.

FR-009 Device Discovery

The Monitoring Agent shall detect:

- New devices
- Removed devices
- Unauthorized devices

FR-010 Traffic Statistics

The system shall calculate:

- Packet rate
- Bandwidth
- Protocol usage
- Source IP distribution

FR-011 Suspicious Traffic Detection

The Monitoring Agent shall identify:

- Abnormally high traffic
- Unknown protocols
- Excessive failed connections

Module 4 – Intrusion Detection Agent

FR-012 Attack Detection

The system shall detect:

- Port scanning
- SQL Injection
- Brute Force
- Malware
- DDoS
- DNS Tunneling

FR-013 Threat Classification

Each detected attack shall receive:

- Attack Category

- Confidence Score
- Risk Level

FR-014 Threat Prioritization

Threats shall be categorized as:

- Critical
- High
- Medium
- Low
- Informational

FR-015 Duplicate Detection

Duplicate alerts shall be merged into one incident.

Module 5 – Threat Correlation Agent

FR-016 Event Correlation

The system shall combine multiple alerts into a single attack sequence.

Example:

FR-017 Timeline Generation

The system shall create attack timelines automatically.

FR-018 Incident Grouping

Related alerts shall be grouped into a single incident.

Module 6 – Knowledge Base Agent

FR-019 Rule Storage

The system shall store:

- Detection Rules
- Firewall Rules
- Security Policies

FR-020 Rule Versioning

Each rule shall maintain:

- Version
- Author
- Date Created
- Last Updated

FR-021 Rule Search

Users shall search rules using:

- Attack Name
- Severity
- Keywords

FR-022 Knowledge Update

The Knowledge Base Agent shall automatically update rules after analyst approval.

Module 7 – Reasoning Agent

FR-023 Forward Chaining

The Reasoning Agent shall support Forward Chaining.

FR-024 Backward Chaining

The system shall verify attack hypotheses using Backward Chaining.

FR-025 Resolution

The system shall resolve logical conflicts.

FR-026 Propositional Logic

The system shall evaluate propositional rules.

FR-027 First Order Logic

The system shall support reasoning over entities including:

- Users
- Devices
- IP Addresses
- Sessions
- Malware

Module 8 – Planning Agent

FR-028 Response Planning

The Planning Agent shall generate multiple response options.

Example:

- Block IP
- Disable User
- Isolate Device
- Ignore

FR-029 Response Ranking

Each response shall receive:

- Cost
- Risk
- Confidence
- Expected Effectiveness

FR-030 Search Algorithms

Planning shall support:

- A*
- Hill Climbing
- Local Search

FR-031 Game-Theoretic Defense

The Planning Agent shall simulate attacker vs. defender strategies to evaluate defensive responses.

Module 9 – Response Agent

FR-032 Firewall Update

The system shall add firewall rules automatically when approved.

FR-033 Endpoint Isolation

The system shall isolate compromised endpoints from the network.

FR-034 User Lockout

The system shall disable compromised user accounts.

FR-035 Alert Notification

Alerts shall be sent via:

- Dashboard
- Email
- SMS (optional)
- Slack or Microsoft Teams (optional)

FR-036 Audit Logging

Every automated response shall be logged.

Module 10 – Learning Agent

FR-037 Incident Storage

Every incident shall be archived.

FR-038 False Positive Learning

Analyst feedback shall improve future detection accuracy.

FR-039 Response Evaluation

The Learning Agent shall measure:

- Response Time
- Success Rate
- Analyst Approval Rate

FR-040 Knowledge Expansion

The Learning Agent shall recommend new detection rules based on historical patterns.

Module 11 – Reporting

FR-041 Incident Report

Generate reports containing:

- Timeline
- Evidence
- Agent Actions
- Final Decision

FR-042 Daily Security Report

Automatically generate daily summaries.

FR-043 Weekly Report

Summarize:

- Threat trends
- Top attack categories
- Agent performance

FR-044 Export Reports

Support:

- PDF
- CSV
- Excel

Module 12 – Logging

FR-045 Activity Logs

Record:

- User Activity
- Agent Activity
- System Events

FR-046 Error Logging

Log all application exceptions.

FR-047 API Logs

Log every API request.

FR-048 Database Logs

Track all database modifications.

Module 13 – Administration

FR-049 Agent Configuration

Administrators shall:

- Enable agents
- Disable agents
- Restart agents
- Update agent settings

FR-050 System Configuration

Configure:

- Detection thresholds
- Time windows
- Alert policies

- Response strategies

---

14. Non-Functional Requirements

Non-functional requirements define **how well** the system performs.

Performance Requirements

NFR-001 Response Time

Threat detection shall complete within **3 seconds** under normal load.

NFR-002 Dashboard Refresh

Dashboard updates shall occur every **2 seconds**.

NFR-003 Throughput

The system shall process at least:

- 1,000 security events per minute (prototype target)
- Scalable to higher rates in future deployments

NFR-004 Concurrent Users

Support at least:

- 100 concurrent users.

Availability

NFR-005 Uptime

Target system availability:

99%.

NFR-006 Recovery

Failed agents shall restart automatically.

Reliability

NFR-007 Fault Tolerance

Failure of one agent shall not stop the entire platform.

NFR-008 Message Retry

Failed messages shall be retried automatically.

Security

NFR-009 Encryption

All API communication shall use HTTPS/TLS.

NFR-010 Password Storage

Passwords shall use secure hashing (e.g., Argon2 or bcrypt).

NFR-011 Authorization

All API endpoints shall require authentication unless explicitly public.

NFR-012 Audit Logging

Every security-sensitive action shall be logged.

Scalability

NFR-013 Horizontal Scalability

New AI agents shall be deployable without redesigning the system.

NFR-014 Database Scalability

The database shall support growth in users, incidents, and logs through indexing and partitioning where appropriate.

Maintainability

NFR-015 Modular Architecture

Each AI agent shall be independently deployable and testable.

NFR-016 Documentation

All public APIs shall be documented (e.g., OpenAPI/Swagger).

NFR-017 Code Quality

Code shall follow established style guides and include unit tests.

Usability

NFR-018 Dashboard

The interface shall be intuitive and require minimal training.

NFR-019 Accessibility

The UI should follow common accessibility guidelines (keyboard navigation, readable contrast, semantic labels).

Compatibility

NFR-020 Platform Support

The application shall run on:

- Windows
- Linux
- macOS (development)

Portability

NFR-021 Containerization

The application shall support Docker deployment.

Explainability

NFR-022 Decision Transparency

Every AI decision shall include:

- Evidence
- Rules Applied
- Confidence Score
- Recommended Action
- Responsible Agent

Compliance

NFR-023 Data Privacy

Sensitive information shall be handled according to applicable organizational policies and relevant data protection regulations.

Backup

NFR-024 Automatic Backup

The system shall support scheduled backups of:

- Incident Database
- Rule Database
- Configuration Files

Disaster Recovery

NFR-025 Recovery Objective

After failure:

- Restore service within **30 minutes** (prototype objective)
- No critical data loss for completed transactions

Requirement Traceability Matrix (Sample)

Requirement ID	Module	AI Unit
FR-008	Monitoring Agent	Unit I
FR-012	Intrusion Detection	Unit II
FR-023	Reasoning Agent	Unit IV
FR-031	Planning Agent	Unit III
FR-032	Response Agent	Units II & III
FR-040	Learning Agent	Units I & IV

PART 4

15. System Design

15.1 Architectural Overview

The Multi-Agent Cyber Defense System follows a **layered, service-oriented architecture** with autonomous agents collaborating through a central orchestration layer.

Architecture Principles

- Modular design
- Agent autonomy
- Loose coupling
- High cohesion
- Event-driven communication
- Explainable AI
- Scalability

High-Level Architecture

---

16. Use Case Diagram

Purpose

Illustrates interactions between users and the system.

Actors

- Administrator
- Security Analyst
- Network Administrator
- AI Agents

Administrator Use Cases

- Login
- Manage Users

- Configure Agents
- Configure Rules
- View Dashboard
- View Reports
- Restart Agents
- Configure System

Security Analyst

- Review Alerts
- Investigate Incidents
- Approve Responses
- Search Knowledge Base
- Export Reports

Network Administrator

- Configure Firewall
- View Network Devices
- Monitor Traffic
- Review Logs

AI Agent

- Monitor Network
- Detect Threats
- Query Knowledge Base
- Execute Reasoning
- Plan Response
- Execute Response

UML Representation

---

17. Class Diagram

Core Classes

User

Attributes

Methods

Agent

Attributes

Methods

Incident

Attributes

Methods

Rule

Attributes

Methods

Threat

Attributes

Methods

Response

Attributes

Methods

Class Relationships

---

18. Component Diagram

The project consists of multiple software components.

---

19. Deployment Diagram

---

20. Sequence Diagram

Attack Detection Sequence

Detailed Sequence

---

21. Data Flow Diagram

Level 0

Level 1

Level 2

---

22. Database Design

Main Tables

Users

Field	Type
user_id	UUID
username	VARCHAR
password_hash	VARCHAR
email	VARCHAR
role	VARCHAR
created_at	TIMESTAMP

Agents

Field	Type
agent_id	UUID
name	VARCHAR
type	VARCHAR
status	VARCHAR
heartbeat	TIMESTAMP

Threats

Field	Type
threat_id	UUID
attack_type	VARCHAR
confidence	FLOAT
severity	VARCHAR
source_ip	VARCHAR
timestamp	TIMESTAMP

Incidents

Field	Type
incident_id	UUID
threat_id	UUID
status	VARCHAR
assigned_to	UUID

Field	Type
created_at	TIMESTAMP

Rules

Field	Type
rule_id	UUID
condition	TEXT
action	TEXT
version	INTEGER
enabled	BOOLEAN

Responses

Field	Type
response_id	UUID
incident_id	UUID
action	VARCHAR
executed_by	VARCHAR
status	VARCHAR
executed_at	TIMESTAMP

Audit Logs

Field	Type
log_id	UUID
user_id	UUID
event_type	VARCHAR
description	TEXT
created_at	TIMESTAMP

---

23. Entity Relationship (ER) Model

---

24. REST API Specification

Authentication

POST `/api/auth/login`

Request

Response

Threat Detection

GET `/api/threats`

Returns all detected threats.

GET `/api/threats/{id}`

Returns detailed information about a specific threat.

Incidents

GET `/api/incidents`

Retrieve incident list.

POST `/api/incidents/{id}/respond`

Execute the planned response for a selected incident.

Rules

GET `/api/rules`

Retrieve all detection rules.

POST `/api/rules`

Create a new rule.

PUT `/api/rules/{id}`

Update an existing rule.

DELETE `/api/rules/{id}`

Disable or remove a rule (subject to authorization policy).

---

25. User Interface Requirements

Dashboard Widgets

The dashboard shall include:

1. Threat Summary Panel
2. Live Network Activity Graph
3. Active Agent Status
4. Recent Incidents
5. Threat Severity Distribution
6. System Health Indicators
7. Agent Communication Log
8. Firewall Action History
9. Alert Timeline
10. AI Decision Explanation Panel

Navigation Menu

- Dashboard
- Live Monitoring
- Threats
- Incidents
- Knowledge Base
- Rules
- Agents
- Reports
- Audit Logs
- Settings
- User Management

PART 5

26. AI Algorithm Design and Implementation

26.1 Introduction

The Multi-Agent Cyber Defense System combines classical Artificial Intelligence algorithms from the academic curriculum with modern Agentic AI principles.

Each algorithm is selected based on the problem-solving requirement of individual agents.

The system does not use AI as a single prediction model. Instead, AI techniques are distributed across multiple autonomous agents responsible for perception, reasoning, planning, and execution.

26.2 Intelligent Agent Model

Agent Definition

An intelligent agent is an autonomous entity that:

1. Observes its environment.
2. Processes available information.
3. Makes rational decisions.
4. Executes actions to achieve goals.

The proposed system follows the PEAS model.

PEAS Representation

Performance Measure

The system evaluates:

- Detection accuracy
- Response time
- False positive rate
- Successful mitigation rate
- System availability

Environment

The agent environment consists of:

- Network traffic
- Users
- Devices
- Applications
- Security logs
- Attack events

Actuators

Agents perform actions:

- Block IP addresses
- Generate alerts
- Update firewall rules
- Isolate machines
- Modify security policies

Sensors

Agents receive information from:

- Packet captures
- Logs
- APIs
- Threat intelligence feeds

---

27. Search Algorithm Integration

27.1 Breadth First Search (BFS)

Usage

Used by the Network Monitoring Agent for network topology exploration.

Example:

Finding reachable devices:

Purpose

- Network discovery
- Device mapping
- Connectivity analysis

27.2 Depth First Search (DFS)

Usage

Used for attack path exploration.

Example:

The agent explores possible attack paths recursively.

27.3 A* Search Algorithm

Usage

Implemented in the Planning Agent.

Purpose:

Find the optimal response strategy.

The algorithm evaluates:

Example:

Possible actions:

Action	Cost
Monitor	Low
Block IP	Medium
Shutdown Server	High

The agent selects the action with maximum security benefit and minimum operational impact.

27.4 Greedy Best First Search

Usage

Fast threat investigation.

The algorithm selects the most promising next step based on threat priority.

Example:

Critical ransomware alert:

27.5 Hill Climbing Search

Usage

Security policy optimization.

Example:

Improving firewall rules:

Initial Rule Set

↓

Evaluate Security Score

↓

Modify Rules

↓

Keep Better Configuration

27.6 Simulated Annealing

Usage

Used when multiple security solutions exist.

Example:

Finding the best combination of:

- Firewall rules
- Monitoring levels
- Access policies

The algorithm avoids local optimum decisions.

---

28. Game Theory Integration

28.1 Attacker vs Defender Model

Cybersecurity can be represented as a strategic game.

Two players:

Attacker

Goal:

- Gain unauthorized access
- Steal information
- Disrupt services

Defender

Goal:

- Protect assets
- Minimize damage
- Restore services

Game Representation

28.2 Alpha-Beta Search

Usage

Response Planning Agent.

The system evaluates possible attacker actions and defensive responses.

Example:

Attacker:

Possible defenses:

The algorithm removes unnecessary branches and selects efficient responses.

28.3 Monte Carlo Tree Search

Usage

Long-term attack prediction.

The agent simulates thousands of possible attack scenarios.

Example:

---

29. Constraint Satisfaction Problem (CSP)

Purpose

Security policies often contain multiple constraints.

Example:

Firewall rule optimization.

CSP Components

Variables

Security rules

Domains

Possible values:

- Allow
- Deny
- Monitor

Constraints

Examples:

- Critical servers cannot expose database ports.
- External users cannot access internal resources.

CSP Example

---

30. Knowledge Representation and Logical Reasoning

30.1 Knowledge Base Structure

The system maintains cybersecurity knowledge:

30.2 Propositional Logic Example

Statement:

Rule:

30.3 First Order Logic Example

Entities:

Example:

30.4 Forward Chaining Workflow

Example:

30.5 Backward Chaining Workflow

Example:

Goal:

Check:

---

31. Cybersecurity Threat Model

31.1 Threat Categories

The system detects:

Malware Attacks

Examples:

- Ransomware
- Trojans
- Spyware

Detection:

- File behavior
- Process monitoring
- Network communication

Network Attacks

Examples:

- DDoS

- Port scanning
- MITM

Detection:

- Traffic analysis
- Packet inspection

Authentication Attacks

Examples:

- Brute force
- Credential stuffing

Detection:

- Login patterns
- User behavior

Insider Threats

Detection:

- Abnormal access
- Data movement
- Privilege misuse

---

32. MITRE ATT&CK Mapping

The system can optionally map detected attacks to MITRE techniques.

Example:

Attack Behavior	Technique
Password Guessing	Credential Attack
Network Scanning	Discovery
Malware Execution	Execution
Data Theft	Exfiltration

PART 6

33. Technology Stack

33.1 Frontend

React.js

Purpose:

- Interactive dashboard
- Real-time visualization
- User interface

33.2 Backend

Python FastAPI

Purpose:

- API services
- Agent communication
- Authentication
- Business logic

33.3 AI Framework

Recommended:

- LangGraph
- CrewAI
- AutoGen

Purpose:

- Agent orchestration
- Workflow management
- Agent memory

33.4 Database

PostgreSQL

Stores:

- Users
- Threats
- Incidents
- Rules
- Logs

33.5 Message Communication

Redis / RabbitMQ

Purpose:

- Agent communication
- Event queues
- Asynchronous processing

33.6 Cybersecurity Tools

Scapy

Used for:

- Packet capture
- Network analysis

Suricata / Zeek

Used for:

- Intrusion detection
- Network monitoring

33.7 Deployment

Docker

Used for:

- Containerization
- Agent isolation
- Easy deployment

---

34. Development Methodology

Agile Development Approach

The project follows an iterative development methodology.

Sprint 1

Requirements and Design

Deliverables:

- PRD
- Architecture
- Database design

Sprint 2

Core Backend

Deliverables:

- Authentication
- APIs
- Database

Sprint 3

Agent Development

Deliverables:

- Monitoring Agent
- Detection Agent
- Knowledge Agent

Sprint 4

Reasoning and Planning

Deliverables:

- Rule engine
- Search algorithms
- Decision system

Sprint 5

Response Automation

Deliverables:

- Firewall actions
- Notifications
- Reports

Sprint 6

Testing and Deployment

Deliverables:

- Security testing
- Performance testing
- Final demonstration

---

35. Testing Strategy

35.1 Testing Levels

Unit Testing

Tests individual components.

Examples:

- Agent functions
- Rule evaluation
- API endpoints

Integration Testing

Tests communication between:

- Monitoring Agent
- Detection Agent
- Reasoning Agent

System Testing

Validates complete workflow.

Example:

Security Testing

Includes:

- Vulnerability scanning
- Authentication testing
- Access control testing

Performance Testing

Measures:

- Detection latency
- Event processing rate
- Database performance

---

36. Sample Test Cases

ID	Test Case	Expected Result
TC001	User login with valid credentials	Login successful
TC002	Invalid password	Access denied
TC003	Port scan detected	Alert generated
TC004	Ransomware behavior detected	Device isolation suggested
TC005	Firewall action executed	Rule updated
TC006	Agent failure occurs	Agent restarted

---

37. Risk Analysis

Risk	Impact	Mitigation
False positives	Medium	Improve reasoning rules
Agent failure	High	Fault recovery mechanism
Data overload	Medium	Event filtering
Incorrect automated action	High	Approval workflow
Security breach	High	Encryption and authentication

---

38. Deployment Architecture

---

39. Future Enhancements

39.1 Reinforcement Learning

Agents can learn optimal defensive strategies through experience.

39.2 Large Language Model Security Analyst

Future versions can include an AI assistant capable of answering:

"Why was this machine isolated?"

"What evidence indicates ransomware?"

39.3 Federated Threat Intelligence

Multiple organizations can share threat knowledge without sharing sensitive data.

39.4 Autonomous Penetration Testing

An attacker simulation agent can continuously test organizational defenses.

39.5 Cloud Security Integration

Future support:

- AWS security monitoring
- Azure security
- Google Cloud security

---

40. Project Success Criteria

The project will be considered successful when:

- Multiple AI agents operate autonomously.
- Threats are detected correctly.
- Agents communicate successfully.
- Logical reasoning produces explanations.
- Automated responses can be executed.
- Dashboard displays real-time security information.
- AI syllabus concepts are demonstrated practically.

41. Research Contribution and Innovation

41.1 Introduction

The Multi-Agent Cyber Defense System introduces an intelligent cybersecurity framework by combining classical Artificial Intelligence techniques with modern Agentic AI architecture.

The primary innovation of this project is the integration of multiple autonomous agents that collaboratively perform cybersecurity operations rather than depending on a single centralized detection model.

The system demonstrates how AI concepts traditionally studied in academic environments can be applied to real-world cybersecurity challenges.

41.2 Research Contributions

Contribution 1: Autonomous Multi-Agent Security Architecture

The project proposes a distributed cybersecurity architecture where independent AI agents cooperate to perform security operations.

Unlike traditional systems:

Traditional Security:

Proposed System:

Benefits:

- Reduced human dependency
- Faster response
- Better scalability
- Improved automation

Contribution 2: Explainable AI-Based Cyber Defense

A major challenge in AI security systems is lack of transparency.

The proposed system provides explanations by recording:

- Evidence collected
- Rules applied
- Logical conclusions
- Decision path
- Selected response

Example:

Contribution 3: AI-Based Security Planning

Most cybersecurity tools only detect attacks.

This system extends beyond detection by planning responses.

The Planning Agent evaluates:

- Threat severity
- Possible actions
- Business impact
- Expected success rate

Algorithms used:

- A* Search
- Game Theory
- Alpha-Beta Search
- Monte Carlo Tree Search

Contribution 4: Knowledge-Driven Cybersecurity

The system uses a cybersecurity knowledge base containing:

- Attack patterns
- Security rules
- Historical incidents
- Response strategies

This enables continuous improvement.

---

42. Academic Mapping and Learning Outcomes

Artificial Intelligence Course Outcomes

The project demonstrates:

AI Concept	Implementation
Intelligent Agents	Autonomous cybersecurity agents
Rational Agents	Security decision-making
Agent Environment	Network environment
Problem Solving Agents	Threat investigation
Search Algorithms	Attack path analysis
Heuristic Search	Response planning
Local Search	Policy optimization
Game Theory	Attacker-defender simulation
CSP	Firewall rule optimization
Knowledge-Based Agents	Security reasoning
Propositional Logic	Threat rules
First Order Logic	Entity-based reasoning
Forward Chaining	Threat inference
Backward Chaining	Investigation workflow
Resolution	Logical conflict solving

---

43. Expected Project Outcomes

After successful implementation, the system will provide:

43.1 Autonomous Threat Detection

The system will automatically identify suspicious activities without requiring continuous manual monitoring.

Expected outputs:

- Threat category
- Severity level
- Confidence score
- Evidence

43.2 Automated Incident Investigation

The system will reconstruct attack sequences.

Example:

43.3 Intelligent Response Recommendation

The system will suggest the most appropriate mitigation action.

Example:

Detected:

Possible Responses:

The Planning Agent selects the best option.

43.4 Improved Security Visibility

The dashboard provides:

- Real-time alerts
- Threat statistics
- Agent activities
- Incident history
- Security reports

---

44. Limitations

Although the proposed system provides autonomous cybersecurity capabilities, certain limitations exist.

44.1 Prototype Scale

The system is designed as an academic and research prototype.

Large enterprise-scale deployment requires additional optimization.

44.2 Dataset Dependency

Detection accuracy depends on:

- Quality of security rules
- Training data
- Threat intelligence sources

44.3 Automated Response Risks

Incorrect automated responses may impact legitimate users.

Therefore, production systems should include:

- Human approval workflows
- Confidence thresholds
- Rollback mechanisms

44.4 Advanced Attack Complexity

Highly sophisticated attacks may require additional:

- Threat intelligence integration
- Behavioral analytics
- Deep learning models

---

45. Security Considerations

The cybersecurity platform itself must be protected.

45.1 Agent Security

Agents must implement:

- Authentication
- Authorization
- Secure communication
- Integrity verification

45.2 Data Protection

Security data must be protected through:

- Encryption
- Access control
- Secure storage
- Audit logging

45.3 API Security

APIs must implement:

- Token-based authentication
- Rate limiting
- Input validation
- Request monitoring

45.4 Model Security

AI components should be protected against:

- Data poisoning
- Prompt injection (for LLM-based extensions)
- Model manipulation

---

46. Project Timeline

6 Month Development Plan

Month	Activities
Month 1	Requirement analysis, literature survey, architecture design

Month	Activities
Month 2	Backend development, database design, authentication
Month 3	Agent development and communication framework
Month 4	AI algorithms, reasoning engine, planning module
Month 5	Dashboard, testing, integration
Month 6	Deployment, documentation, presentation

---

47. Project Team Responsibilities

Team Member 1: AI Architecture

Responsibilities:

- Agent design
- Search algorithms
- Reasoning engine

Team Member 2: Cybersecurity Module

Responsibilities:

- Threat detection
- Network monitoring
- Attack simulation

Team Member 3: Backend Development

Responsibilities:

- APIs
- Database
- Authentication

Team Member 4: Frontend Development

Responsibilities:

- Dashboard

- Visualization
- Reports

---

48. Demonstration Scenario

Scenario 1: Brute Force Attack

Step 1

Attacker attempts multiple login attempts.

Step 2

Monitoring Agent observes abnormal authentication activity.

Step 3

Detection Agent calculates threat score.

Step 4

Reasoning Agent applies rules:

Step 5

Planning Agent evaluates responses.

Step 6

Response Agent blocks attacker IP.

Step 7

Dashboard displays:

---

49. Final Conclusion

The Multi-Agent Cyber Defense System using Agentic AI presents a next-generation approach to cybersecurity automation by combining autonomous intelligent agents with classical Artificial Intelligence algorithms.

The project demonstrates that concepts such as intelligent agents, heuristic search, game theory, constraint satisfaction, and logical reasoning can be practically applied to solve complex cybersecurity problems.

The proposed architecture enables multiple AI agents to collaborate, analyze threats, reason about security events, plan optimal responses, and execute defensive actions.

Unlike traditional security systems that primarily detect threats, this system moves toward autonomous cyber defense by introducing intelligent decision-making capabilities.

The project provides:

- A practical implementation of Artificial Intelligence concepts.
- A scalable cybersecurity architecture.
- Explainable security decisions.
- Automated threat response.
- A foundation for future autonomous security systems.

The proposed solution represents an important step toward building intelligent cybersecurity platforms capable of adapting to continuously evolving digital threats.

---

50. References

Books

1. Stuart Russell and Peter Norvig,
   Artificial Intelligence: A Modern Approach,
   4th Edition, Pearson.
2. William Stallings,
   Network Security Essentials: Applications and Standards,
   Pearson.
3. Ross Anderson,
   Security Engineering: A Guide to Building Dependable Distributed Systems.

Research Areas

1. Multi-Agent Systems in Cybersecurity.
2. Autonomous Security Operations Centers.
3. Explainable Artificial Intelligence for Cyber Defense.
4. AI-Based Intrusion Detection Systems.
5. Game Theory Applications in Network Security.

Technologies

- Python Documentation
- FastAPI Documentation
- PostgreSQL Documentation
- Docker Documentation
- React Documentation
- Scapy Documentation

---

51. Appendix A – Sample Security Rules

Rule 001: Brute Force Detection

Rule 002: Ransomware Detection

---

52. Appendix B – Agent Message Format

---

53. Appendix C – Glossary

Term	Meaning
Agentic AI	AI systems capable of autonomous planning and execution
Agent	Autonomous software entity capable of decision making
IDS	Intrusion Detection System
CSP	Constraint Satisfaction Problem
FOL	First Order Logic

Term	Meaning
SIEM	Security Information and Event Management
SOC	Security Operations Center
Threat Intelligence	Information about cyber threats
Heuristic Search	Search using estimated future cost
Knowledge Base	Repository of facts and rules